CREALITY V4.2.2 / ENDER-3 V2 — ALL CONNECTORS

22 keyed callouts: 20 connector/terminal interfaces + 2 unpopulated pad groups

RESEARCH REFERENCE • callout numbers are NOT contact numbers • no wiring or firmware changes

07 X MOTOR
4-pin coil output

08 Y MOTOR
4-pin coil output

09 Z MOTOR
4-pin coil output

10 E MOTOR
4-pin coil output

01 24 V DC INPUT
2 screws • fixed DC supply

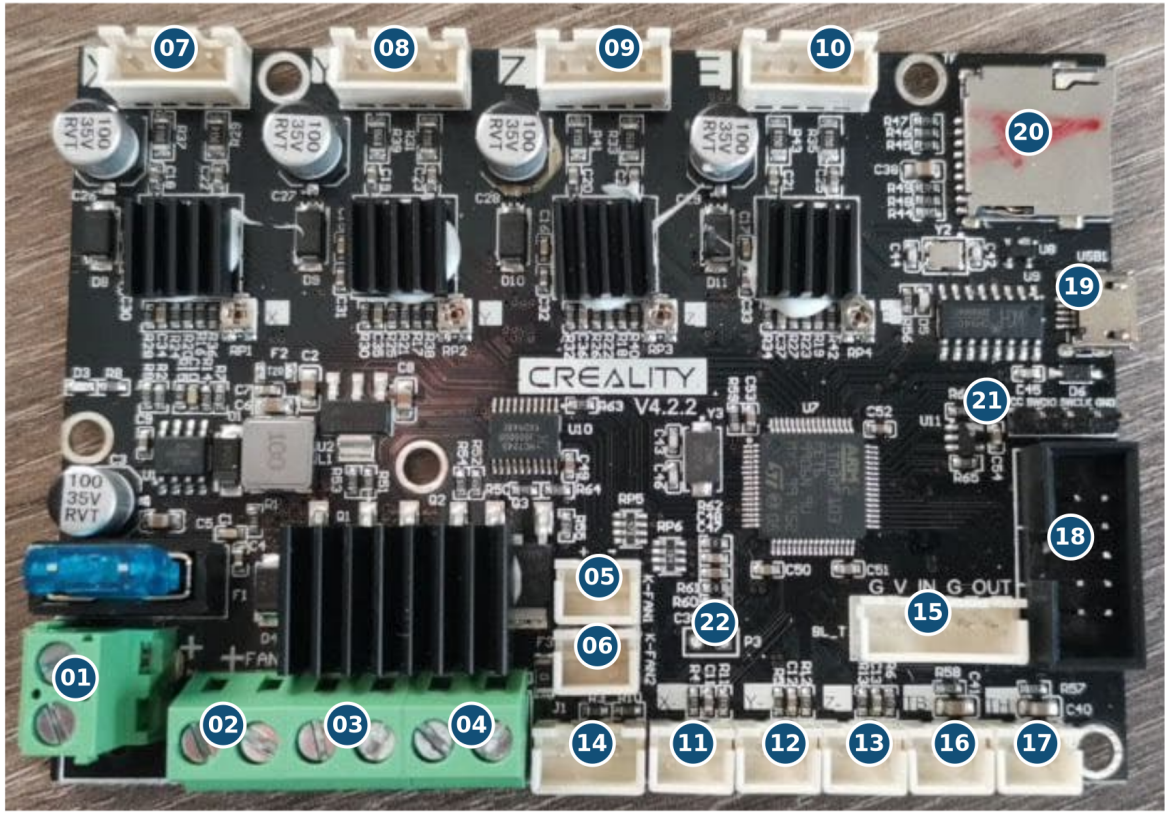
02 FIXED FAN + / -
2 screws • always-on 24 V

03 BED HEATER
2 screws • switched-power output

04 NOZZLE HEATER
2 screws • switched-power output

05 K-FAN1
2-pin • controlled 24 V power

06 K-FAN2
2-pin • shares K-FAN1 control



18 DISPLAY / EXP
10-pin IDC • UART + controls
Occupied; not spare PWM

19 MICRO-USB
USB-to-serial bridge
Not raw MCU PWM

20 MICROSD
SDIO storage interface
Not spare PWM

21 P2 / SWD PADS
4 unpopulated debug pads
3.3 V / PA13 / PA14 / GND

22 P3 AUX PADS
2 pads • PC0 + GND
"Reset" label is misleading

11 X- ENDSTOP
2-pin input • PA5

12 Y- ENDSTOP
2-pin input • PA6

13 Z- ENDSTOP
2-pin input • PA7

14 J1 RUNOUT
3-pin input • PA4

15 BL_T PROBE
5-pin • PB0 / PB1 • occupied

16 TB THERMISTOR
2-pin analog input • PC4

17 TH THERMISTOR
2-pin analog input • PC5

PWM: power outputs ≠ TTL; endstop/runout inputs ≠ clean PWM; motor coils ≠ signal supplies.

Reference photo: ender3.club • V4.2.2 silk checked • full electrical details, caveats and sources accompany this image.

Power and fan connections

Fixed supply and low-side-switched 24 V outputs are electrically different from logic PWM.

01 24 V IN | 2 screw terminals

Function / electrical: Main DC supply input. 24 V DC, positive and board ground.
MCU: None. **PWM:** NO: supply input.
Limits / evidence: Not mains. Physical screw polarity must follow the actual board markings. [S1, S2]

02 FAN + / - (fixed-power pair) | 2 screw terminals

Function / electrical: Always-powered hotend/heatsink fan supply. 24 V DC; permanent ground return.
MCU: None. **PWM:** NO: fixed supply.
Limits / evidence: Not K-FAN1/2. Fan fuse/rail routing differs between published drawings and manufacturing reports; do not assume which fuse protects it. [S1, S2]

03 BED + / - | 2 screw terminals

Function / electrical: Heated-bed power output. 24 V positive; MOSFET-switched negative.
MCU: PA2 -> buffer -> bed MOSFET. **PWM:** POWER PWM / heater switching; not TTL.
Limits / evidence: PA2 supports native timer channels at the MCU, but those are not exposed here. Negative terminal is not permanent ground. [S1, S3, S4]

04 NOZ + / - (hotend heater) | 2 screw terminals

Function / electrical: Original nozzle-heater output; current test laser-enable output. 24 V positive; MOSFET-switched negative.
MCU: PA1 -> buffer -> nozzle MOSFET. **PWM:** POWER PWM capable; current laser test uses ON/OFF enable.
Limits / evidence: PA1 is not raw logic at the screws. Common ground tied to its switched negative can bypass the power switch. No heater or laser connection recommended. [S1, S3, S4; current test context]

05 K-FAN1 | 2-pin socket

Function / electrical: Controlled fan output. 24 V positive; shared low-side MOSFET return.
MCU: PA0 -> buffer -> fan MOSFET. **PWM:** POWER PWM, not TTL.
Limits / evidence: Shares one control/net with K-FAN2: not independent channels. Marlin declares FAN_SOFT_PWM_REQUIRED. Current SCOPE6 context reports about 7.8 Hz software PWM, not laser-frequency PWM. [S1, S3; current test context]

06 K-FAN2 | 2-pin socket

Function / electrical: Second controlled fan output (often board-cooling fan). 24 V positive; same switched return as K-FAN1.
MCU: PA0, shared with K-FAN1. **PWM:** POWER PWM, not TTL.
Limits / evidence: Same command/duty as K-FAN1; not a separate spare output. Do not confuse it with the fixed FAN screw pair. [S1, S3]

Motor and endstop connectors

Four motor sockets plus three TWO-PIN endstop inputs. Motor coils are never TTL signal contacts.

07 X MOTOR | 4-pin socket

Function / electrical: X stepper-motor coil connections. Two current-regulated H-bridge coil pairs; approximately 24 V switching, not a DC supply.
MCU: PC2 STEP, PB9 DIR; PC3 shared ENABLE (upstream of driver). **PWM:** NO usable logic PWM at socket; internal driver chopper switching is NOT a laser signal.
Limits / evidence: MCU STEP/DIR pins are not the motor socket contacts. Driver type varies by V4.2.2 manufacturing run. All four enables share PC3. [S1, S3]

08 Y MOTOR | 4-pin socket

Function / electrical: Y stepper-motor coil connections. Two current-regulated H-bridge coil pairs; approximately 24 V switching, not a DC supply.
MCU: PB8 STEP, PB7 DIR; PC3 shared ENABLE (upstream of driver). **PWM:** NO usable logic PWM at socket; internal driver chopper switching is NOT a laser signal.
Limits / evidence: MCU STEP/DIR pins are not the motor socket contacts. Driver type varies by V4.2.2 manufacturing run. All four enables share PC3. [S1, S3]

09 Z MOTOR | 4-pin socket

Function / electrical: Z stepper-motor coil connections. Two current-regulated H-bridge coil pairs; approximately 24 V switching, not a DC supply.
MCU: PB6 STEP, PB5 DIR; PC3 shared ENABLE (upstream of driver). **PWM:** NO usable logic PWM at socket; internal driver chopper switching is NOT a laser signal.
Limits / evidence: MCU STEP/DIR pins are not the motor socket contacts. Driver type varies by V4.2.2 manufacturing run. All four enables share PC3. [S1, S3]

10 E MOTOR | 4-pin socket

Function / electrical: E stepper-motor coil connections. Two current-regulated H-bridge coil pairs; approximately 24 V switching, not a DC supply.
MCU: PB4 STEP, PB3 DIR; PC3 shared ENABLE (upstream of driver). **PWM:** NO usable logic PWM at socket; internal driver chopper switching is NOT a laser signal.
Limits / evidence: MCU STEP/DIR pins are not the motor socket contacts. Driver type varies by V4.2.2 manufacturing run. All four enables share PC3. [S1, S3]

11 X- ENDSTOP | 2-pin socket

Function / electrical: X minimum limit-switch input. Signal plus ground; 3.3 V input network, NO supply contact.
MCU: PA5. **PWM:** INPUT as wired; not a clean logic-PWM output.
Limits / evidence: R11=10 kohm series, R4=10 kohm pull-up to 3.3 V, C11 to ground (100 nF ASSUMED by schematic author). No native timer PWM on PA5. Divider can hold a driven-low socket near 1.65 V. [S1, S3, S4]

12 Y- ENDSTOP | 2-pin socket

Function / electrical: Y minimum limit-switch input. Signal plus ground; 3.3 V input network, NO supply contact.
MCU: PA6. **PWM:** INPUT as wired; not a clean logic-PWM output.
Limits / evidence: R12=10 kohm series, R5=10 kohm pull-up to 3.3 V, C12 to ground (100 nF ASSUMED by schematic author). PA6 can carry TIM3_CH1 at the MCU. Divider can hold a driven-low socket near 1.65 V. [S1, S3, S4]

Motor and endstop connectors (continued)

Four motor sockets plus three TWO-PIN endstop inputs. Motor coils are never TTL signal contacts.

13 Z- ENDSTOP | 2-pin socket

Function / electrical: Z minimum limit-switch input. Signal plus ground; 3.3 V input network, NO supply contact.
MCU: PA7. **PWM:** INPUT as wired; not a clean logic-PWM output.
Limits / evidence: R13=10 kohm series, R6=10 kohm pull-up to 3.3 V, C13 to ground (100 nF ASSUMED by schematic author). PA7 can carry TIM3_CH2 / TIM8_CH1N at the MCU. Divider can hold a driven-low socket near 1.65 V. User measured about 3.3 V HIGH / 1.6 V LOW and rounded ~844 mVpp at ~1 kHz; this is user-reported, not a new measurement. [S1, S3, S4; Z measurements supplied by user]

Runout, probe and temperature inputs

The working 3D Touch stays connected. Sensor-supply discrepancies and physical orientation remain explicit.

14 J1 / filament runout | 3-pin socket

Function / electrical: Optional filament detector. PA4 signal, ground, sensor-supply contact.
MCU: PA4. **PWM:** INPUT as wired; PA4 has NO native timer-PWM alternate function.
Limits / evidence: R10=10 kohm series; R3=10 kohm pull-up; no signal capacitor shown. V4.2.2 schematic/wiki specify 3.3 V supply, but a combined TH3D page illustrating V4.2.7 says 5 V. Actual rail unmeasured. No outer-contact physical order asserted. [S1, S3, S4, S5, S6]

15 BL_T / probe (G V IN G OUT) | 5-pin socket

Function / electrical: 3D Touch / BLTouch-compatible probe power, deploy and trigger. Two grounds, 5 V supply, 3.3 V MCU control/trigger nets.
MCU: PB0 deploy/control; PB1 trigger. **PWM:** SERVO control pulses on PB0; PB1 is trigger INPUT.
Limits / evidence: Occupied by working 3D Touch: KEEP. PB0/PB1 have timer alternate functions, but are not spare laser pins. No series/endstop RC network shown in these probe signal paths; not permission to attach a laser. [S1, S3, S4, S5; current wiring context]

16 TB / bed thermistor | 2-pin socket

Function / electrical: Bed-temperature sensor input. Analog input plus ground, 3.3 V pull-up/filter network.
MCU: PC4. **PWM:** NO native timer PWM; analog sensing input.
Limits / evidence: Reference: R58=4.7 kohm pull-up, C41 labelled 10 uF. Not a power output. Sensor connector positions can be mislabelled in marketing images; PCB silk and trace take precedence. [S1, S3, S4]

17 TH / hotend thermistor | 2-pin socket

Function / electrical: Nozzle-temperature sensor input. Analog input plus ground, 3.3 V pull-up/filter network.
MCU: PC5. **PWM:** NO native timer PWM; analog sensing input.
Limits / evidence: Reference: R57=4.7 kohm pull-up, C40 labelled 10 uF. Not a power output. Capacitance values are reference-drawing values, not measurements of this board. [S1, S3, S4]

Laser-specific conclusion: This document does not approve a direct Snapmaker connection to any socket. The Snapmaker's reported active-low, approximately 4 V control interface still needs a verified interface design. The Z socket divider/filter and the heater switched-return bypass problem are not solved by changing connector names.

Display, storage and service interfaces

Unpopulated service pads are labelled as pads, not imaginary fitted sockets.

18 Display / EXP (IDC ribbon) | 10-pin shrouded socket

Function / electrical: Ender-3 V2 display serial link, encoder/button/beeper signals. 3.3 V MCU signals plus 5 V display supply and ground.
MCU: RET6 map: PC6, PB2, PB10, PB11, PB14, PB13, PB12, PB15. **PWM:** NOT a stock spare PWM connector; some nets have timer alternate functions.
Limits / evidence: PB10/PB11 are USART3; display uses other nets for controls/beeper. Schematic includes 47-ohm series networks. Occupied display signals cannot simply be reassigned. No numbered ribbon-pin orientation given here. [S1, S3, S5]

19 USB1 / micro-USB | 5 USB contacts plus shell

Function / electrical: Computer serial connection through CH340G bridge. USB differential D+/D-, VBUS, ground; not a raw MCU USB/PWM header.
MCU: CH340G -> MCU PA9 TX / PA10 RX (USART1). **PWM:** NO: USB/serial interface.
Limits / evidence: USB can partially power logic; do not assume board is dead merely because 24 V is off. USB D+/D- do not expose PA11/PA12; those MCU pins serve the onboard EEPROM here. [S1, S3]

20 microSD socket | 8 card contacts plus card-detect switch

Function / electrical: Firmware/print-file storage. 3.3 V SDIO bus.
MCU: PC8..PC11 data; PC12 clock; PD2 command; PC7 detect. **PWM:** NO usable stock PWM interface.
Limits / evidence: Storage bus contacts are not spare PWM outputs. SDIO uses these nets; PA4 is runout, not this socket chip-select. [S1, S3]

21 P2 / SWD programming pads | 4 pads; header not fitted in reference photo

Function / electrical: Debug/programming interface. 3.3 V reference, SWDIO, SWCLK, ground.
MCU: PA13 SWDIO; PA14 SWCLK. **PWM:** NO native timer PWM on PA13/PA14.
Limits / evidence: Schematic calls this ICSP; on STM32 it is SWD, not AVR ICSP. Pads, not a fitted four-pin socket. Do not assume the voltage pad is 5 V. [S1, S3, S4; photo P2 silk]

22 P3 auxiliary pads ("Reset" in drawing) | 2 pads; header not fitted

Function / electrical: Auxiliary pulled-up input/test-pad pair. PC0 signal plus ground; 3.3 V pull-up/filter.
MCU: PC0 (MCU pin 8) in traced circuit, NOT NRST pin 7. **PWM:** NO native timer PWM on PC0.
Limits / evidence: Drawing labels it P3 Reset, but its traced net and author correction identify PC0. R60=10 kohm pull-up, R61=0 ohm link, C39 to ground (value assumed). Treat reset naming cautiously; no shorting instructions. [S1, S2, S4; photo P3 silk]

Evidence, confidence and audit

Read together with the numbered photograph and the complete Markdown/CSV inventory.

Scope: STM32F103RET6-based Ender-3 V2 V4.2.2 reference, visibly revision-checked. Component, driver and fan-fuse variants exist. This is not a factory schematic and not a fresh photograph of Michael's individual board.

Inventory audit: 20 connector/terminal interfaces plus two unpopulated service-pad groups are identified. The fixed FAN screw pair is separate from both K-FAN sockets. X/Y/Z endstops are two-pin. No separate fitted reset switch or external I2C socket was identified. The EEPROM, fuses, LEDs, ICs, resistor networks and crystals are components, not additional connectors.

Confidence: High: physical connector identities/counts and common MCU net assignments. Reference-level: resistor/capacitor values, rail routing and individual variant matching. J1 supply remains unmeasured. Capacitors marked 100 nF inherit the schematic author's assumption, not a measurement.

Pin numbering: All 01-22 labels are connector callout IDs, NOT contact numbers. No physical left/right polarity is inferred from a schematic symbol. Probe silk is reproduced as a name only, not a new wiring instruction.

Firmware distinction: Hardware capability is not the currently configured output. Servo pulses, software fan PWM, heater power switching, timer alternate functions and stepper-driver chopping are separate classifications. SCOPE6 observations and Z measurements are supplied context, not newly verified hardware tests.

S1 — Community traced V4.2.2 schematic, 28-5-22; NOT an official Creality schematic
https://github.com/Jyers/Marlin/files/8791332/Creality.4.2.2.-.Schematic.28-5-22.pdf

S2 — Schematic author discussion/corrections, including PC0 at P3 and grounded MOSFET sources
https://github.com/Jyers/Marlin/discussions/814

S3 — Marlin upstream V4 pin map; RET6 configuration, not V4.2.7 overrides
https://github.com/MarlinFirmware/Marlin/blob/bugfix-2.1.x/Marlin/src/pins/stm32f1/pins_CREALITY_V4.h

S4 — ST STM32F103xC/D/E manufacturer datasheet, alternate-function Table 5
https://www.st.com/resource/en/datasheet/stm32f103re.pdf

S5 — Independent V4.2.2 community photo and J1/probe/display documentation
https://ender3.club/Creality_4.2.2

S6 — Combined TH3D 4.2.2/4.2.7 article: illustrates 4.2.7; conflicting J1 supply claim
https://tickets.th3dstudio.com/products/article/creality-v4-2-7-v4-2-2-wiring-and-pinout-diagram

Photo — Original photograph visibly labelled V4.2.2; reference board, not a newly captured photo of Michael's unit
https://ender3.club/images/3/34/Pinount_creality_4.2.2.png